

Non-Binding Futility — Validation of the MAMS / MAMSS Changes

Purpose. We added non-binding futility support to MAMS (boundary derivation) and evaluated the resulting designs with MAMSS. This document shows that the three designs behave exactly as theory predicts — FWER is controlled, power is met, and expected sample size orders correctly — for **both** stopping methods (**simultaneous** and **sep**).

Three metrics per config (MAMSS, R = 25,000 simulated trials):

- **FWER** = $P(\text{reject} \geq 1 \text{ arm} \mid H_0) - \text{target} \leq \alpha = \mathbf{0.05}$
- **Power** = $P(\text{reject} \geq 1 \text{ arm} \mid H_1) - \text{target} \geq \mathbf{0.90}$
- **ESS** = expected total sample size (reported under H1 and H0)
- **CSP** = $P(\text{reject the effective arm AND it has the largest statistic} \mid H_1) - \text{“correct-selection” power (did we pick the *right* arm?)}$. For K=1, CSP = Power.

The three designs

Folder	Efficacy boundary	Futility in simulation	What its numbers mean
binding	binding	enforced (<code>fut.arm.simple</code>)	realistic FWER, power, ESS
non-binding	non-binding (wider)	NOT acted on (<code>fut.arm.nostop</code>)	worst-case FWER ; ESS overstated
non-binding-enforced	non-binding (wider)	enforced (<code>fut.arm.simple</code>)	realistic power & ESS; conservative FWER

The non-binding boundary must control FWER *even if the trial team never acts on the futility boundary* — that worst case is the **non-binding** (nostop) run. The **non-binding-enforced** run shows the realistic power/ESS seen when the futility boundary *is* honoured.

Stopping methods (context)

simultaneous applies one shared stopping rule: the whole trial stops as soon as any arm crosses efficacy (or all remaining arms cross futility). **sep** (“separate”) gives each arm its own group-sequential test, so one arm’s win or drop does not stop the others — the trial runs until every arm is individually resolved. Both methods use the **same** boundaries returned by `mams()`.

Note: in the **non-binding** designs the futility boundary is never acted on, so there the trial can only stop for **efficacy** (futility neither drops arms nor ends the trial).

Config grid (identical in all three folders)

$K \in \{1, 4\} \cdot J \in \{1, 2\} \cdot p \in \{0.65, 0.75\} \cdot p0 = 0.5 \cdot \alpha = 0.05 \cdot \text{power} = 0.90 \cdot \text{method} \in \{\text{simultaneous}, \text{sep}\} \rightarrow 2 \times 2 \times 2 \times 2 = \mathbf{16 \text{ configs per folder}}$.
Boundary shapes: `ushape = "obf"`, `lshape = "triangular"` (see note below).

Run 2026-06-22 · MAMS 3.0.3 with the `binding` argument · MAMSS. One PDF per method per folder: `<folder>/pdf/initial-simultaneous.pdf` and `initial-sep.pdf` (text summary + size plot per config).

Note — why `lshape = "triangular"`

The non-binding correction only changes the boundary when the futility bound is **aggressive** enough that ignoring it actually inflates α . With OBF (interim bound ~ -3.07) almost no H_0 path crosses it, so non-binding is a no-op — identical boundaries to binding. Triangular puts the interim bound at $1[1] \approx +0.76$ ($K=4, J=2$): ignoring it leaves $\sim 78\%$ of H_0 paths in play, which inflates α , so u must widen. That widening is exactly what the comparison needs to test, which is why the grid uses triangular. Seen at the design level ($J=2$ configs):

config	u binding	u non-binding	Max N binding	Max N non-binding
K=1, J=2	2.294, 1.622	2.373, 1.678	128	132
K=4, J=2	3.025, 2.139	3.072, 2.172	420	430

($J=1$ configs have no interim, so binding and non-binding coincide there.)

What we expect (the assumptions to verify)

Check	Condition	Applies to
A	$\text{FWER} \leq 0.05$ in every config	all three designs (worst case = nostop)
B	$\text{Power} \geq 0.90$	binding, non-binding-enforced (futility honoured)
C	$\text{ESS}(\text{nostop}) \geq$ $\text{ESS}(\text{enforced}) \geq$ $\text{ESS}(\text{binding})$	non-binding pair vs binding

Check	Condition	Applies to
D	Max N(non-binding) $\geq$ Max N(binding)	non-binding vs binding
E	sep and simultaneous: same FWER & Power , sep ESS_H1 $\geq$ simultaneous	both methods

Why E holds: sep and simultaneous share the boundary, and `reject` ≥ 1 is triggered by the *same* first efficacy crossing — so FWER and Power are identical. They differ only in **ESS under H1**, where a winning arm under **sep** no longer halts the others, so the trial enrolls more. (K=1 has nothing to separate; J=1 has no interim — identical there.)

Monte Carlo 95% band (R = 25,000 simulated trials per config), used for the approximate tolerance below. With an estimated rate `x` from `R` trials, the band is $1.96 \cdot \text{sqrt}(x(1-x)/R)$:

FWER : $1.96 \cdot \text{sqrt}(0.05 \cdot 0.95 / 25000) = 1.96 \cdot 0.00138 = 0.0027$ (+/-)
Power: $1.96 \cdot \text{sqrt}(0.90 \cdot 0.10 / 25000) = 1.96 \cdot 0.00190 = 0.0037$ (+/-)

Results

Tables below are the **simultaneous** runs. **sep** is identical on FWER, Power, ESS_H0 and Max N; it differs only on ESS_H1 (summarised in the sep table at the end; full per-config numbers in `initial-sep.pdf`).

Binding

*Focus: **FWER** (≤ 0.05) and **Power** (≥ 0.90) — the realistic reference design.*

K	J	p	FWER	Power	ESS H1	ESS H0	Max N
1	1	0.65	0.0486	0.9017	116.0	116.0	116
1	1	0.75	0.0505	0.9004	38.0	38.0	38
1	2	0.65	0.0488	0.8994	95.2	81.6	128
1	2	0.75	0.0474	0.9130	32.5	28.0	44
4	1	0.65	0.0485	0.9000	400.0	400.0	400
4	1	0.75	0.0502	0.9131	135.0	135.0	135
4	2	0.65	0.0492	0.8995	278.4	267.4	420
4	2	0.75	0.0478	0.9075	92.3	89.0	140

Non-binding, futility NOT acted on (worst-case FWER)

Focus: **FWER** — the worst case the non-binding boundary must control. (Also **Max N**, for Check D.)

K	J	p	FWER	Power	ESS H1	ESS H0	Max N
1	1	0.65	0.0486	0.9017	116.0	116.0	116
1	1	0.75	0.0505	0.9004	38.0	38.0	38
1	2	0.65	0.0509	0.9270	103.1	131.5	132
1	2	0.75	0.0470	0.9335	34.2	43.8	44
4	1	0.65	0.0485	0.9000	400.0	400.0	400
4	1	0.75	0.0502	0.9131	135.0	135.0	135
4	2	0.65	0.0498	0.9156	367.6	429.3	430
4	2	0.75	0.0480	0.9210	119.7	139.7	140

ESS here is **overstated by design** (`fut.arm.nostop` → no early stopping, ESS $\approx$ Max N). This run’s role is the worst-case FWER, not ESS.

Non-binding, futility enforced

Focus: **Power** (≥ 0.90) and **ESS** — the realistic operating characteristics.

K	J	p	FWER	Power	ESS H1	ESS H0	Max N
1	1	0.65	0.0486	0.9017	116.0	116.0	116
1	1	0.75	0.0505	0.9004	38.0	38.0	38
1	2	0.65	0.0448	0.9006	99.7	83.6	132
1	2	0.75	0.0413	0.9057	33.1	27.9	44
4	1	0.65	0.0485	0.9000	400.0	400.0	400
4	1	0.75	0.0502	0.9131	135.0	135.0	135
4	2	0.65	0.0460	0.8987	285.5	273.1	430
4	2	0.75	0.0442	0.9033	92.9	88.8	140

sep vs simultaneous (K=4, J=2 — the only configs that differ)

FWER and Power are identical to the simultaneous tables above; only ESS_H1 changes:

design	p	FWER (=both)	Power (=both)	ESS_H1 sim	ESS_H1 sep	Δ ESS_H1 (sep-sim)
binding	0.65	0.0492	0.8995	278.4	302.9	+24.5
binding	0.75	0.0478	0.9075	92.3	100.7	+8.4
non-binding (nostop)	0.65	0.0498	0.9156	367.6	417.4	+49.8

design	p	FWER (=both)	Power (=both)	ESS_H1 sim	ESS_H1 sep	Δ ESS_H1 (sep-sim)
non-binding (nostop)	0.75	0.0480	0.9210	119.7	135.9	+16.2
enforced	0.65	0.0460	0.8987	285.5	310.2	+24.7
enforced	0.75	0.0442	0.9033	92.9	100.8	+7.9

(ESS_H0 is ~unchanged: under H0 every arm drops on its own, so separating the stop barely matters — the cost of **sep** is borne under H1.)

Correct-selection power (CSP)

Power above counts *any* rejection. CSP is stricter: the truly effective arm (A) must be both **rejected** and have the **largest test statistic** (i.e. correctly identified as the best arm). With $R = 25,000$ trials, arm A the effective arm, $\text{type}_{a,r} = 1$ if arm a was rejected in trial r , and $T_{a,r}$ its test statistic at its stopping look:

$$\text{CSP} = \frac{1}{R} \sum_{r=1}^R \mathbf{1}[\text{type}_{A,r} = 1] \cdot \mathbf{1}\left[A = \arg \max_{a \in \{1, \dots, K\}} T_{a,r}\right]$$

i.e. the fraction of trials in which A is both rejected and the largest statistic. Reported power, by contrast, drops the selection term and only asks whether *some* arm was rejected: $\text{Power} = \frac{1}{R} \sum_r \mathbf{1}[\exists a : \text{type}_{a,r} = 1]$.

In this grid only arm A is effective, so:

- for **K=1**, CSP = Power exactly (one arm — nothing to select);
- for **K=4**, CSP is marginally below Power — A is the largest statistic in ~99.5% of trials, so correct selection is essentially “free” here. The gap would only widen if several arms were effective with similar effect sizes.

We display the **K=4, J=2** configs because that is the only setting where CSP is a meaningful question *and* the sequential machinery is exercised: K=1 has no arms to choose between (CSP = Power), and J=1 has no interim looks. K=4/J=1 gaps are similar (~0.0004) and K=1 rows are omitted as they equal Power exactly.

design	config (K=4)	Power	CSP	gap (Power – CSP, ≥ 0)
binding	J=2, p=0.65	0.8995	0.8986	0.0009
binding	J=2, p=0.75	0.9075	0.9064	0.0011
non-binding (nostop)	J=2, p=0.65	0.9156	0.9151	0.0005
non-binding (nostop)	J=2, p=0.75	0.9210	0.9202	0.0008
enforced	J=2, p=0.65	0.8987	0.8981	0.0006

design	config (K=4)	Power	CSP	gap (Power – CSP, ≥ 0)
enforced	J=2, p=0.75	0.9033	0.9024	0.0009

The gap is **non-negative by construction**: the CSP event (A rejected *and* A largest) is a subset of the Power event (≥ 1 arm rejected), evaluated on the same trials — so $\text{CSP} \leq \text{Power}$ always, and the gap is a “selection loss” whose magnitude (not sign) is what matters.

(Computed from the per-trial statistics stored with `extended = 1`; for `simultaneous` all arms are compared at the shared stopping look. `sep` gives essentially the same CSP.)

Takeaway: rejection almost always coincides with correctly selecting the effective arm, so the non-binding change preserves not just power but correct selection.

Checks (results vs the conditions above)

- ☒ **A — FWER controlled** (≤ 0.05 , every config). Max FWER: binding 0.0505, nostop 0.0509, enforced 0.0505 — all within the ± 0.0027 band. **PASS.** Enforced FWER is visibly lower on J=2 (e.g. 0.0460, 0.0413): enforcing futility removes paths that could cross u , so it is conservative — hence the worst-case check uses nostop.
- ☒ **B — Power ≥ 0.90** (futility honoured). Min power: binding 0.8994, enforced 0.8987 — within the ± 0.0037 band. **PASS.** (nostop power is inflated to 0.92–0.93 because it never stops; enforced lands on ~ 0.90 , confirming n is correctly sized.)
- ☒ **C — ESS ordering** nostop $\geq$ enforced $\geq$ binding (J=2). e.g. K=4, J=2, p=0.65 (H0): $429.3 \geq 273.1 \geq 267.4$. **PASS.** (binding vs enforced differ only at the ~ 0.1 level where Max N is equal after rounding = MC noise.)
- ☒ **D — Max N(non-binding) $\geq$ Max N(binding)**. K=1, J=2: $132 \geq 128$; K=4, J=2: $430 \geq 420$ (J=1 / equal-after-rounding configs tie, but u is still wider). **PASS.**
- ☒ **E — sep vs simultaneous.** FWER and Power identical in every config; sep ESS_H1 strictly larger on every K=4/J=2 config (see table), equal where K=1 or J=1. **PASS.**

Conclusion

All five checks pass, so the non-binding implementation behaves as theory requires for both stopping methods. The non-binding boundary controls FWER even in the worst case where the futility boundary is ignored, still hits the target power once futility is honoured, and the `sep` / `simultaneous` choice moves

only expected sample size (under H_1) — never the error rate or power. The non-binding option is therefore safe to use and behaves as documented.